BERT Aspect Sentiment Analysis

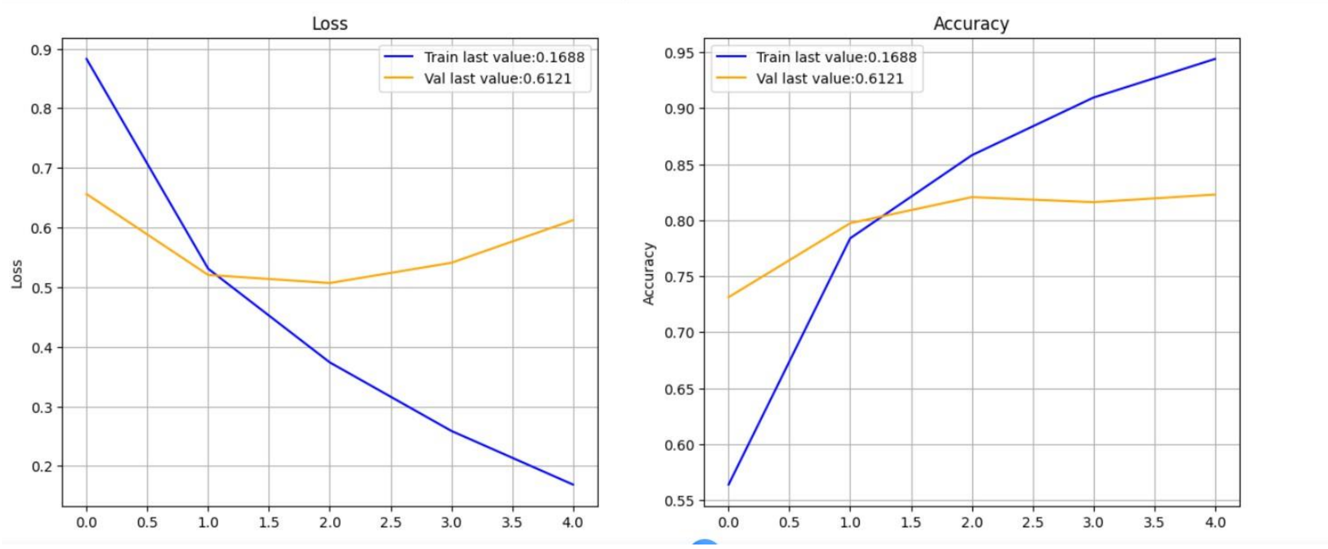
Question 1

Basic classifiers using BERT: Model 1 and Model 2. The code for these models is partially provided: complete the missing code, run it, and compare the performance of Model 1 and Model 2. Explain in your report the differences you see.

Model 2 achieved a slightly higher peak validation accuracy than Model 1 but Model 1 achieved better test accuracy overall. This suggests that the prebuilt sequence classification model generalizes better to unseen data. A possible reason could be that Model 1 preserves more task-relevant contextual information while Model 2 compresses token representations through average pooling, which may dilute important aspect-specific sentiment cues. Because averaging treats all tokens more equally and reduces detailed structure. It may blur important sentiment words negations.

Model 1

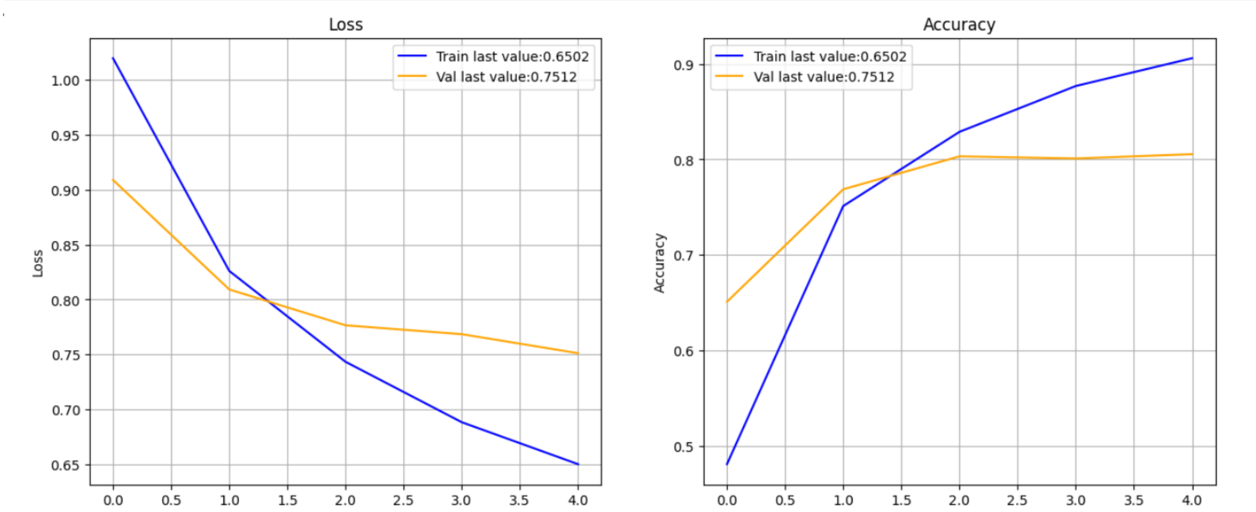
Test Loss: 0.5379, Test Acc: 0.8263



Training loss decreases steadily across epochs, while validation loss initially decreases but begins to increase after the second epoch which indicates mild overfitting.

Model 2:

Test Loss: 0.7449, Test Acc: 0.8226



Unlike Model 1, its validation loss does not rise sharply. So Model 2 seems a bit more stable and slightly less overfit-looking.

Question 2

Advanced classifier using BERT: Model 3. You must construct, train and test a CNN or LSTM model using BERT embeddings in a similar way to Model 2. In your report, compare the performance you get against Task 2 above. Briefly discuss and explain the differences you see.

Model 3:

Test Loss: 0.6527, Test Acc: 0.8211

From the experimental results model 2 achieved a test accuracy of 82.26% while model 3 achieved a test accuracy of 82.11% kind of similar results not much difference.Although the LSTm architecture allows the model to capture word order information but still the improvement is limited.This may be likely because BERT embeddings already encode most of the contextual and sequential information.